



IoT Weather Monitoring System

Group Members:

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Introduction

- Our project focuses on developing an **IoT-based Weather Monitoring System** that leverages the ESP-32 microcontroller and multiple sensors to collect real-time weather data.
- By integrating the DHT11 sensor (temperature and humidity), BMP280 sensor (pressure and altitude), and MQ135 sensor (air quality), we aim to provide a comprehensive environmental monitoring solution.
- The data is transmitted via MQTT to a local MySQL database and visualized using Power BI, ensuring efficient storage, retrieval, and insights for smarter decision-making.

Purpose

Problem Statement:

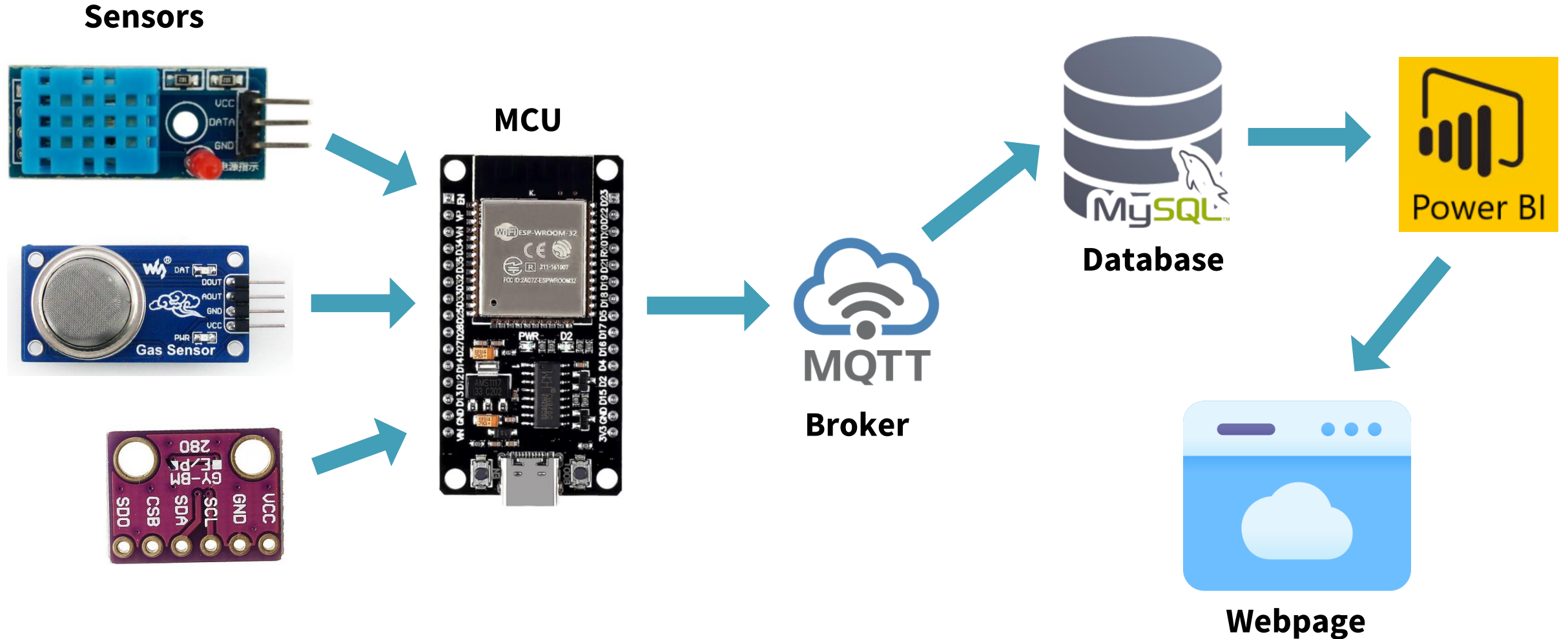
- In older times, weather monitoring systems are often limited by their **lack of real-time data integration** and **inability to combine critical environmental parameters** such as temperature, humidity, pressure, altitude, and air quality.
- Lack accessible visualization tools and **IoT-enabled data transmission**, making them less effective for dynamic weather analysis.

Need: An IoT-enabled, real-time, multi-sensor solution.

Goal/Solution

- This project addresses these gaps by **developing an existing IoT-based solution** that collects, transmits, stores, and visualizes weather data using the ESP-32 platform, MQTT protocol, MySQL database, and Power BI dashboard.

System Block Diagram



Components

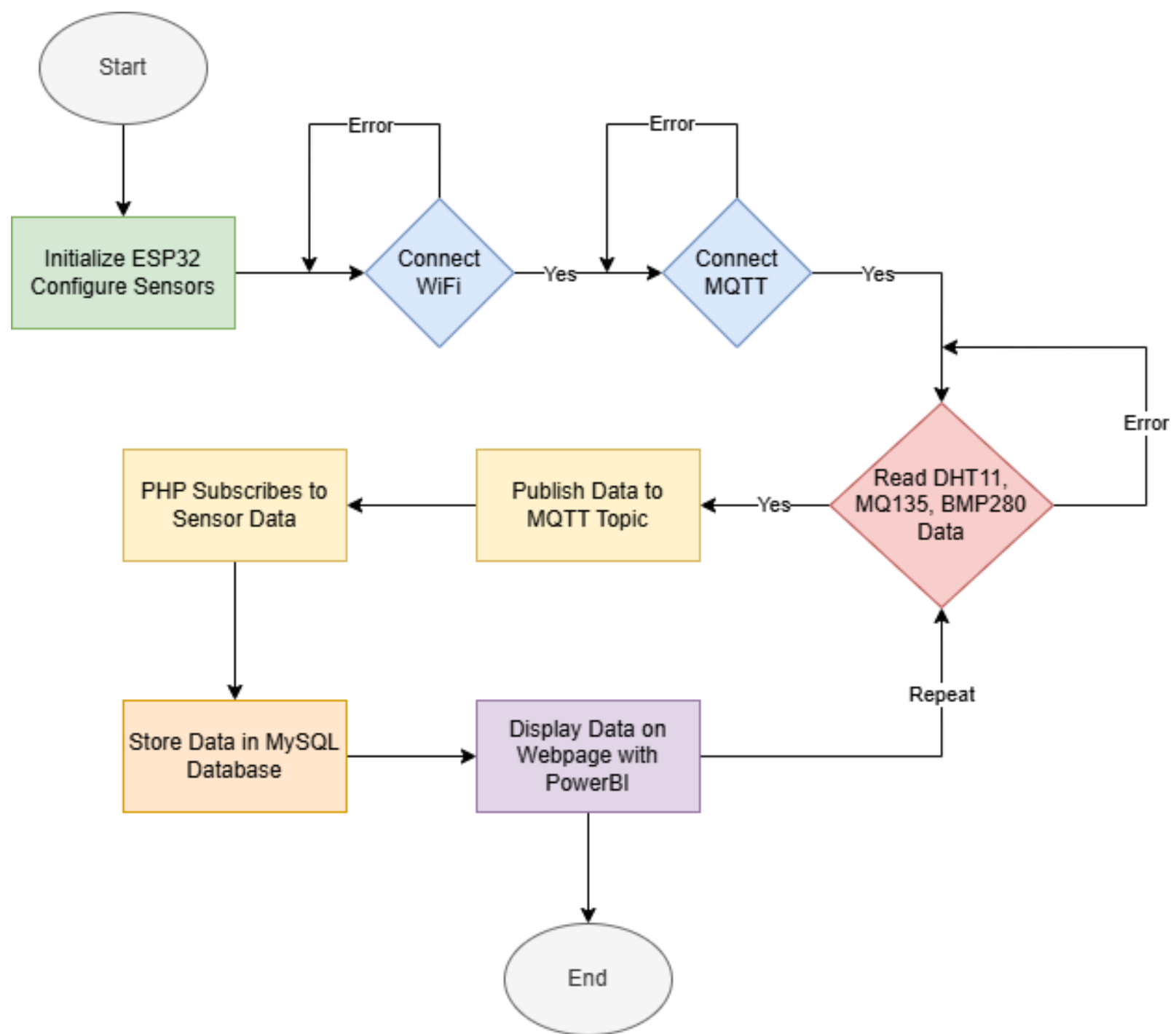
Software Needed:

- Arduino IDE
- EMQX Broker
- XAMPP Web Server Stack
- MySql Database
- PowerBi

Hardware Needed

Item	Quantity	Cost
ESP32 MCU	1	Rs. 1180
DHT11 Sensor (Temperature + Humidity)	1	Rs. 250
MQ135 Sensor (Air Quality)	1	Rs. 340
BMP280 (Pressure + Altitude)	1	Rs. 350
Connector Wires	20+	Already Have
Breadboard	1	Already Have
Laptop	1	Already Have
USB Cable	1	Already Have

Project Flowchart/ Methodology



Implementation

iot_project | Arduino IDE 2.3.3
File Edit Sketch Tools Help

ESP32-WROOM-DA M...

LIBRARY MANAGER

Filter your search...

Type: Installed
Topic: All

Arduino Uno WiFi Dev
Ed Library by Arduino
0.0.3 installed
This library allows users to use network features like rest and mqtt. Includes some tools for...
More info
0.0.3 REMOVE

Adafruit BMP280
Library by Adafruit
2.6.8 installed
Arduino library for BMP280 sensors. Arduino library for BMP280 pressure and altitude...
More info
2.6.8 REMOVE

Adafruit BusIO
by Adafruit
1.16.1 installed

iot_project.ino

```
1 #include <DHT.h>
2 #include <Adafruit_Sensor.h>
3 #include <Adafruit_BMP280.h> // Include BMP280 library
4
5 // DHT11 Sensor Setup
6 #define DHTPIN 27 // DHT11 data pin DAT
7 #define DHTTYPE DHT11 // DHT11 sensor type
8 DHT dht(DHTPIN, DHTTYPE);
9
10 // MQ135 Sensor Setup
11 #define MQ135_AO_PIN 25 // MQ135 Analogue Output pin AO
12 #define MQ135_DO_PIN 26 // MQ135 Digital Output pin DO
13
14 // BMP280 Sensor Setup
15 #define BMP280_I2C_ADDRESS 0x76 // BMP280 I2C address (default is 0x76)
```

Output Serial Monitor x

Message (Enter to send message to 'ESP32-WROOM-DA Module' on 'COM6')

Altitude: 188.52 m
Gas Concentration: 1249 ppm
Gas level exceeds threshold!
Failed to read from DHT11 sensor!
Pressure: 990.82 hPa
Altitude: 188.40 m
Gas Concentration: 1250 ppm
Gas level exceeds threshold!
Failed to read from DHT11 sensor!
Pressure: 990.81 hPa
Altitude: 188.55 m
Gas Concentration: 1255 ppm
Gas level exceeds threshold!

Output Serial Monitor x

Message (Enter to send message to 'ESP32-WROOM-DA Module' on 'COM6')

Gas level exceeds threshold!

Temperature: 25.30 °C

Humidity: 39.00 %

Pressure: 866.43 hPa

Altitude: 1301.11 m

Gas Concentration: 1067 ppm

Gas level exceeds threshold!

Temperature: 25.30 °C

Humidity: 39.00 %

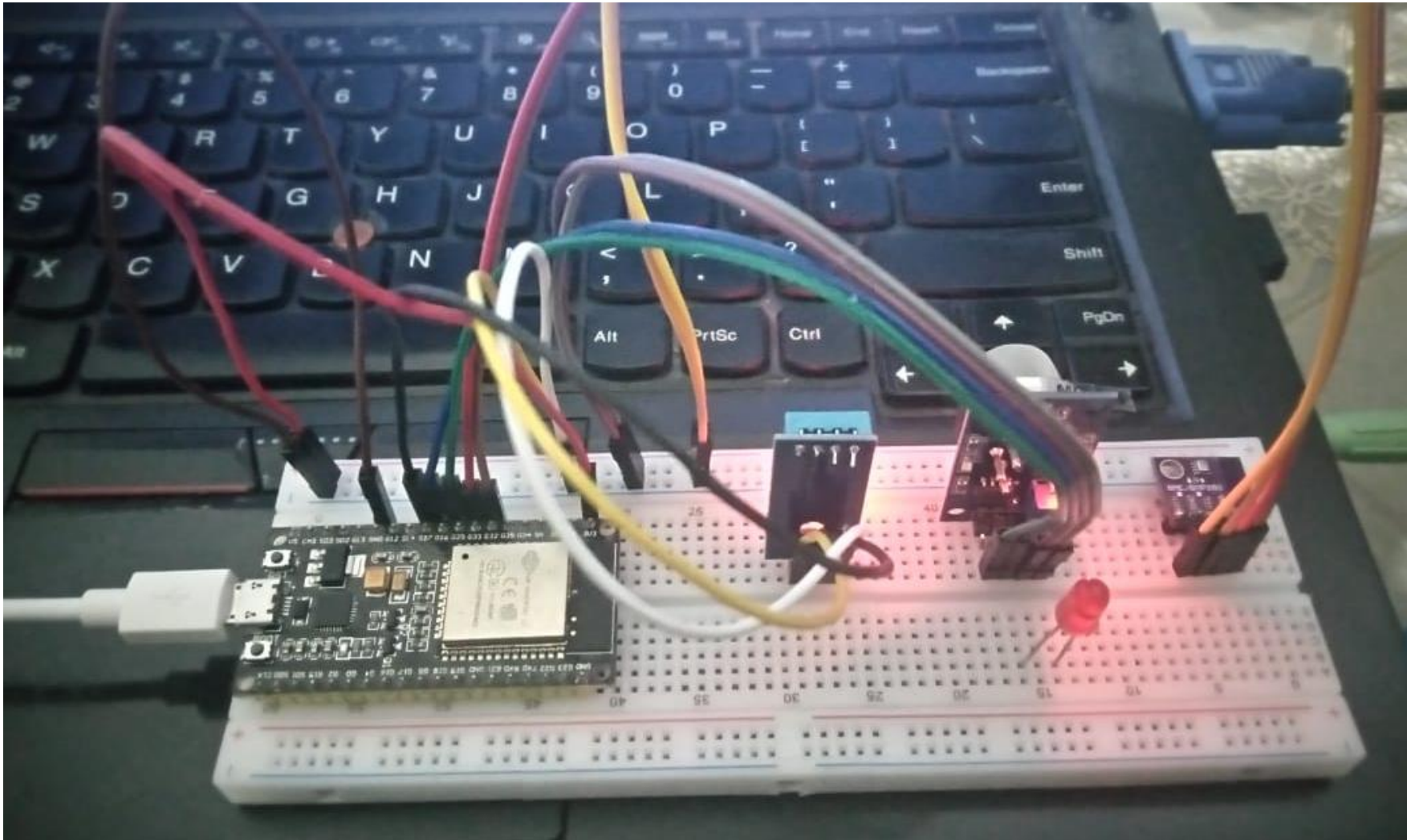
Pressure: 866.43 hPa

Altitude: 1301.11 m

Gas Concentration: 1066 ppm

Gas level exceeds threshold!

Hardware Configuration



Hardware Configuration

```
// DHT11 Sensor Setup
#define DHTPIN 27          // DHT11 data pin DAT
#define DHTTYPE DHT11     // DHT11 sensor type
DHT dht(DHTPIN, DHTTYPE);

// MQ135 Sensor Setup
#define MQ135_AO_PIN 25 // MQ135 Analogue Output pin AO
#define MQ135_DO_PIN 26 // MQ135 Digital Output pin DO

// BMP280 Sensor Setup
#define BMP280_I2C_ADDRESS 0x76 // BMP280 I2C address (c
#define SDA_PIN 32 // BMP280 I2C SDA pin
#define SCL_PIN 33 // BMP280 I2C SCL pin
Adafruit_BMP280 bmp;
```

Output Serial Monitor ×

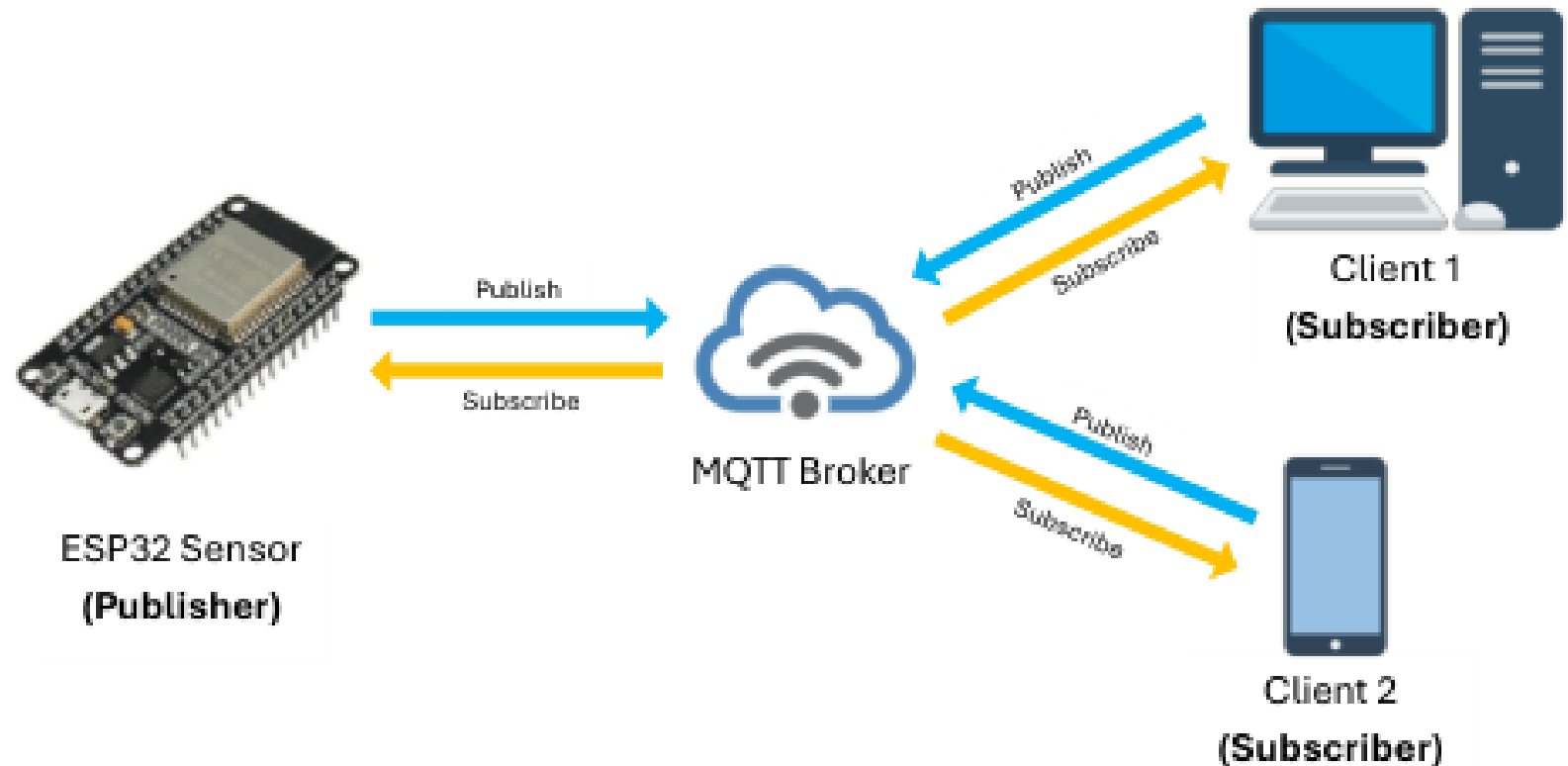
Message (Enter to send message to 'ESP32-WROOM-DA Module' on 'COM6')

```
Gas level exceeds threshold!
Temperature: 25.30 °C
Humidity: 39.00 %
Pressure: 866.43 hPa
Altitude: 1301.11 m
Gas Concentration: 1067 ppm
Gas level exceeds threshold!
Temperature: 25.30 °C
Humidity: 39.00 %
Pressure: 866.43 hPa
Altitude: 1301.11 m
Gas Concentration: 1066 ppm
Gas level exceeds threshold!
```

MQTT Configuration

MQTT Overview

- **MQTT (Message Queuing Telemetry Transport)** is a lightweight client-server messaging protocol designed for efficient data exchange in IoT applications
- It operates on a **publish/subscribe model**, where clients communicate by publishing messages to specific topics and subscribing to topics to receive messages.



MQTT Configuration

```
#include <DHT.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_BMP280.h>
#include <WiFi.h>
#include <PubSubClient.h>

// WiFi Settings
const char* ssid = "FSQUARE";
const char* password = "rochelle";

// MQTT broker settings
const char* mqtt_server = "broker.emqx.io"; // Use EMQX broker
const char* publish_topic = "weather_data"; // Topic to publish to
const char* subscribe_topic = "mqttx_data"; // Topic to subscribe to
```



EMQX

MQTT Configuration

```
void connectToWiFi() {  
    WiFi.begin(ssid, password);  
    Serial.print("Connecting to WiFi...");  
    while (WiFi.status() != WL_CONNECTED) {  
        delay(1000);  
        Serial.print(".");  
    }  
    Serial.println("Connected to WiFi!");  
}
```

```
String createPayload() {  
    // Create a JSON payload  
    String payload = "{";  
    payload += "\"temperature\": " + String(temperature) + ",";  
    payload += "\"humidity\": " + String(humidity) + ",";  
    payload += "\"pressure\": " + String(pressure) + ",";  
    payload += "\"altitude\": " + String(altitude) + ",";  
    payload += "\"gas_value\": " + String(gasValue) + ",";  
    payload += "}";  
    return payload;  
}  
  
void reconnectMQTT() {  
    // Loop until we're reconnected to the MQTT broker  
    while (!client.connected()) {  
        Serial.print("Attempting MQTT connection...");  
        if (client.connect("ESP32Client")) {  
            Serial.println("connected");  
            client.subscribe(subscribe_topic); // Re-subscribe to the  
        } else {  
            Serial.print("failed, rc=");  
            Serial.print(client.state());  
            Serial.println(" try again in 5 seconds");  
            delay(5000);  
        }  
    }  
}
```

MQTT Configuration

MQTTX Interface

ⓘ Edit Subscription

*

Topic

weather_data

*

QoS

0

At most once

Color

#4AD772

Subscribe Topic

MQTTX

File Edit View Window Help

MQTTX

+

</>

Connections

EMQX Test@broker.e...

● **iot_project@broker.e...**

iot_project

✓

87

+ New Subscription

weather_data QoS 0

Plaintext

All

Received

Published

2024-12-09 08:45:15:645

Topic: weather_data QoS: 0

{ "temperature": 24.10, "humidity": 40.00, "pressure": 994.16, "altitude": 160.13, "gas_value": 1072, "gas_detected": false }

2024-12-09 08:45:37:099

Topic: weather_data QoS: 0

{ "temperature": 23.80, "humidity": 40.00, "pressure": 994.15, "altitude": 160.27, "gas_value": 1072, "gas_detected": true }

2024-12-09 08:45:47:146

JSON QoS 0 Retain Meta

mqttx_data

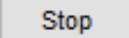
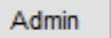
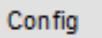
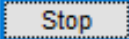
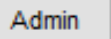
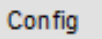
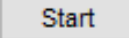
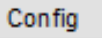
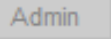
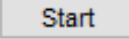
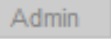
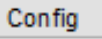
{ "msg": "Hello back from MQTTX" }

Database Setup

XAMPP Control Panel v3.3.0 [Compiled: Apr 6th 2021]



XAMPP Control Panel v3.3.0

Service	Module	PID(s)	Port(s)	Actions
	Apache	2972 22116	80, 443	  
	MySQL	19888	3306	  
	FileZilla			  
	Mercury			  
	Tomcat			  

8:22:54 AM [mysql]

8:22:54 AM [mysql]

8:22:54 AM [mysql]

8:22:54 AM [mysql]

8:22:54 AM [mysql]

8:22:54 AM [mysql]

8:28:01 AM [mysql]

8:28:02 AM [mysql]

This may be due to a blocked port, missing dependencies, improper privileges, a crash, or a shutdown by another method. Press the Logs button to view error logs and check the Windows Event Viewer for more clues. If you need more help, copy and post this entire log window on the forums.

Attempting to start MySQL app...

Status change detected: running



MySQL Connector/ODBC Data Source Configuration

Test Result

Connection successful

OK

Connection Parameters

Data Source Name: IoT_Projec_Server

Description:

☒ TCP/IP Server: Port: 3306

☐ Named Pipe:

User: root

Password:

Database: iot_project

Test

Details >>

OK

Cancel

Help

Database Setup

```
<?php
$servername = "localhost";
$username = "root";
$password = "";
$dbname = "iot_project";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);

// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// Set MySQL timezone to Asia/Karachi
$conn->query("SET time_zone = '+05:00'");
```

Database Setup

```
// Parse the JSON payload
$data = json_decode($rawData, true); // true for associative array

// Check if the required fields are present
if (isset($data['Temperature'], $data['Humidity'], $data['Pressure'], $data['Altitude'], $data['Air_Quality_Index'])) {
    $temperature = $data['Temperature'];
    $humidity = $data['Humidity'];
    $pressure = $data['Pressure'];
    $altitude = $data['Altitude'];
    $gas_value = $data['Air_Quality_Index'];

    // Insert parsed data into the database
    $sql = "INSERT INTO weather_data (`Time`, `Temperature`, `Humidity`, `Pressure`, `Altitude`, `Air Quality Index`)
        VALUES (NOW(), '$temperature', '$humidity', '$pressure', '$altitude', '$gas_value')";

    if ($conn->query($sql) === TRUE) {
        echo "Data inserted successfully from MQTTX simulation";
    } else {
        echo "Error: " . $sql . "<br>" . $conn->error;
    }
} else {
    echo "Invalid JSON payload: Required fields are missing.";
}
```

Database Setup

←T→				ID	Time	Temperature	Humidity	Pressure	Altitude	Air Quality Index
☐	 Edit	 Copy	 Delete	1	2024-12-09 13:58:47	24.8	40	994.02	161.33	1016
☐	 Edit	 Copy	 Delete	2	2024-12-09 13:58:49	24.5	39	994	161.5	1007
☐	 Edit	 Copy	 Delete	3	2024-12-09 13:58:51	24.3	41	994.02	161.33	1014
☐	 Edit	 Copy	 Delete	4	2024-12-09 13:58:54	24.5	40	994.02	161.31	1059
☐	 Edit	 Copy	 Delete	5	2024-12-09 13:58:56	24.5	41	994	161.48	1062
☐	 Edit	 Copy	 Delete	6	2024-12-09 13:58:58	24.8	40	994.03	161.23	1064
☐	 Edit	 Copy	 Delete	7	2024-12-09 13:59:00	24.3	41	994.06	161.04	1002
☐	 Edit	 Copy	 Delete	8	2024-12-09 13:59:02	24.3	41	994.04	161.14	1063
☐	 Edit	 Copy	 Delete	9	2024-12-09 13:59:05	24.3	39	994.04	161.17	1067
☐	 Edit	 Copy	 Delete	10	2024-12-09 13:59:07	24.5	41	994.03	161.27	1030
☐	 Edit	 Copy	 Delete	11	2024-12-09 13:59:09	24.3	40	994.06	161.01	1080
☐	 Edit	 Copy	 Delete	12	2024-12-09 13:59:11	24.3	39	994.05	161.08	1029

Webpage

Weather Database

Time	Temperature (°C)	Humidity (%)	Air Pressure (hPa)	Altitude (m)	Air Quality Index (ppm)
2024-12-09 13:59:11	24.3	39	994.05	161.08	1029
2024-12-09 13:59:09	24.3	40	994.06	161.01	1080
2024-12-09 13:59:07	24.5	41	994.03	161.27	1030
2024-12-09 13:59:05	24.3	39	994.04	161.17	1067
2024-12-09 13:59:02	24.3	41	994.04	161.14	1063
2024-12-09 13:59:00	24.3	41	994.06	161.04	1002
2024-12-09 13:58:58	24.8	40	994.03	161.23	1064
2024-12-09 13:58:56	24.5	41	994	161.48	1062
2024-12-09 13:58:54	24.5	40	994.02	161.31	1059
2024-12-09 13:58:51	24.3	41	994.02	161.33	1014
2024-12-09 13:58:49	24.5	39	994	161.5	1007

PowerBI

Connecting to ODBC MySQL Server

From ODBC

Data source name (DSN)

IoT_Project_Server

Advanced options

Navigator

Display Options

ODBC (dsn=IoT_Project_Server) [7]

information_schema

iot_project [2]

test

☒ weather_data

mysql

performance_schema

phpmyadmin

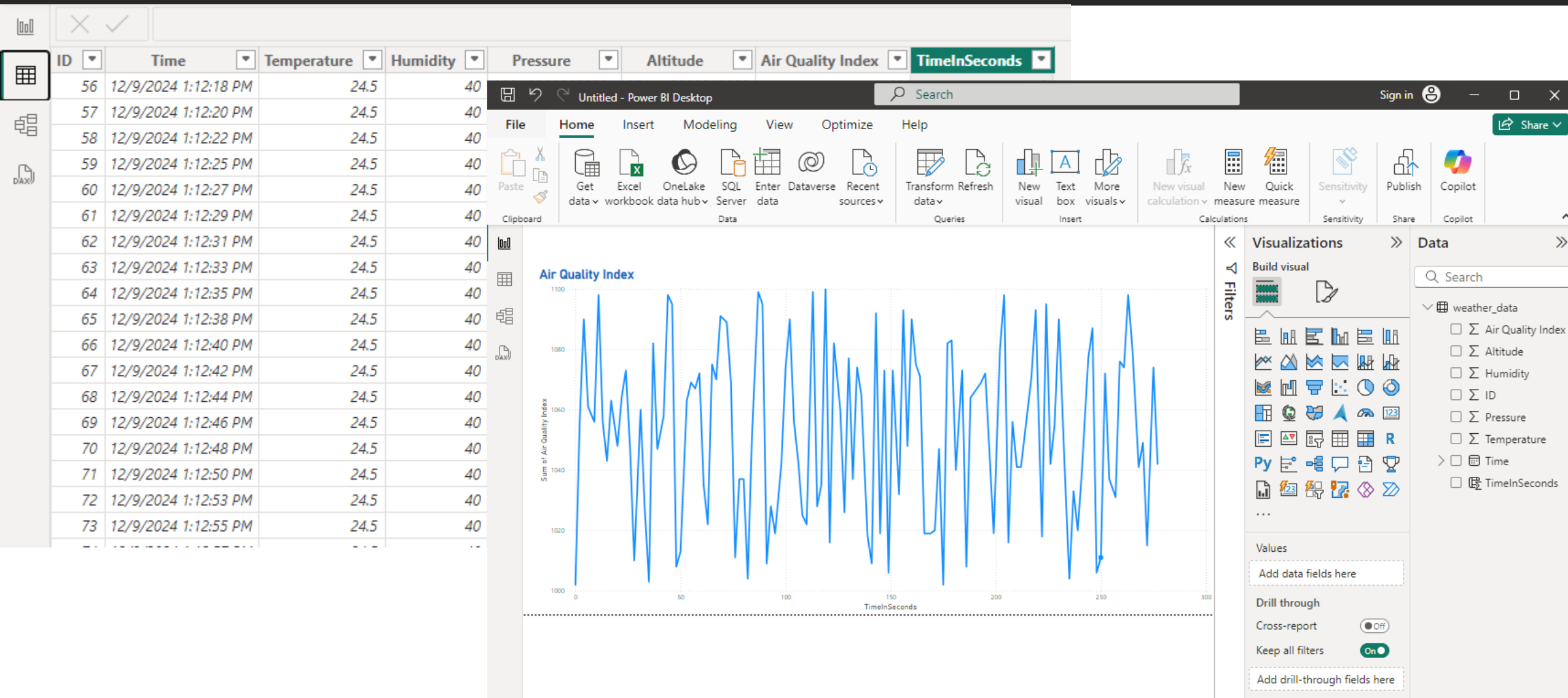
project

test

weather_data

ID	Time	Temperature	Humidity	P
1	12/9/2024 1:10:19 PM	24.5	41	P
2	12/9/2024 1:10:21 PM	24.5	41	
3	12/9/2024 1:10:23 PM	24.5	41	
4	12/9/2024 1:10:25 PM	24.5	41	
5	12/9/2024 1:10:28 PM	24.5	41	
6	12/9/2024 1:10:30 PM	24.5	41	
7	12/9/2024 1:10:32 PM	24.5	41	
8	12/9/2024 1:10:34 PM	24.5	41	
9	12/9/2024 1:10:36 PM	24.5	41	
10	12/9/2024 1:10:39 PM	24.5	41	
11	12/9/2024 1:10:41 PM	24.5	41	
12	12/9/2024 1:10:43 PM	24.5	41	
13	12/9/2024 1:10:45 PM	24.5	41	
14	12/9/2024 1:10:47 PM	24.5	41	
15	12/9/2024 1:10:50 PM	24.5	41	
16	12/9/2024 1:10:52 PM	24.5	41	

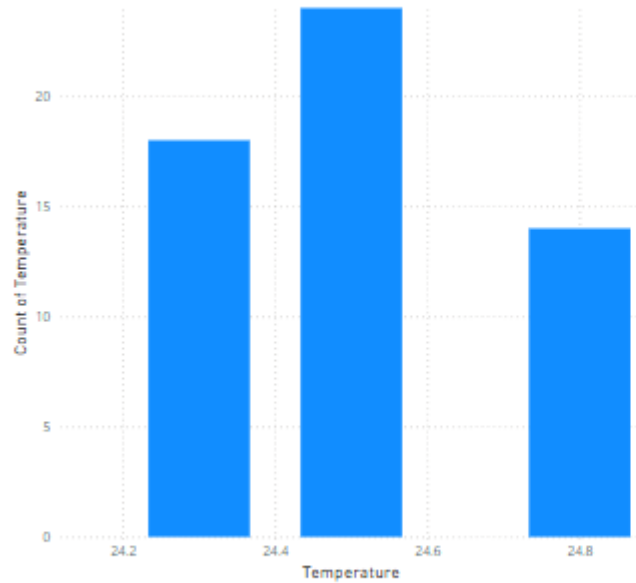
PowerBI



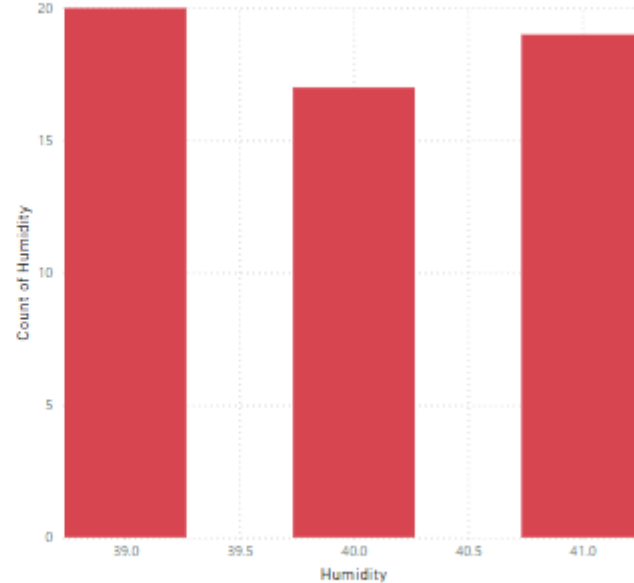
PowerBI



Count of Temperature

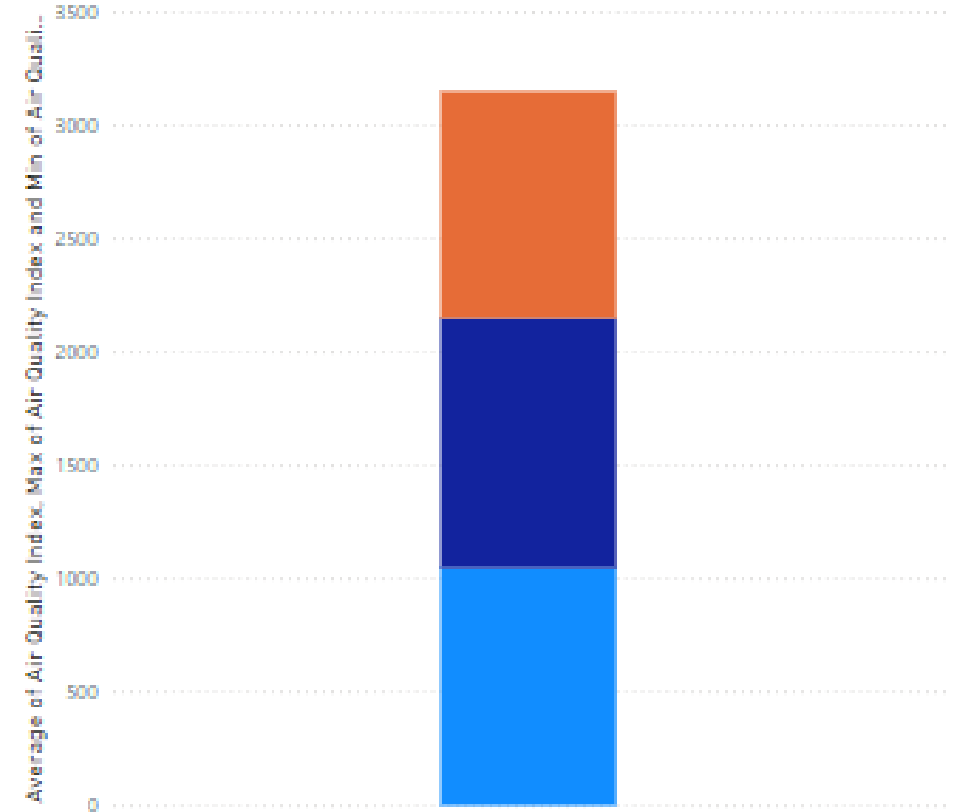


Count of Humidity



Average, Maximum, and Minimum of Air Quality Index

● Average of Air Quality Index ● Max of Air Quality Index ● Min of Air Quality Index



Task Division

- Generally, the project has been a collective team effort, but some individual tasks were assigned by the team leader to each team member.

Awwab	Zafar	Zohaib
• Project Manager • Ordering Components • Hardware Setup • MQTT Configuration • Presentation	• Hardware Setup • Project Research • Troubleshooting • PowerBI Integration • Diagram Making	• Presentation • Database Setup • Webpage Designer • PowerBI Integration

Marks Deserved (Out of 100)

1. Effort & Dedication: (28/30 marks)

- Extensive research and significant time spent on sensors, protocols, and tools.
- Debugging and resolving hardware/software issues.

2. Modules Covered: (36/40 marks)

Sensor Configuration: Setting up DHT11, BMP280, MQ135 (10/10 marks).

MQTT Protocol: Real-time data transmission (8/10 marks).

Database: MySQL integration for data storage (5/5 marks).

Frontend Visualization: Power BI dashboards (8/10 marks).

3. Innovation & Application: (8/10 marks)

- Creative use of IoT technologies for comprehensive weather monitoring.

4. Presentation & Documentation: (13/15 marks)

- Clear and concise explanation of the project in slides.
- Well-documented code and diagrams for system understanding.

90/100

Conclusion

Conclusion

- Our IoT Weather Monitoring System successfully integrates temperature, humidity, air quality, pressure, and altitude monitoring using DHT11, MQ135, and BMP280 sensors with the ESP-32 microcontroller.
- The system enables real-time data transmission via MQTT, efficient storage in a MySQL database, and interactive visualization through Power BI and a frontend, demonstrating a practical solution for real-time environmental monitoring

Future Work

- **Additional Sensors:** Integrate additional sensors for parameters like rainfall and wind speed.
- **Advanced Analytics:** Incorporate machine learning for weather predictions and anomaly detection.
- **Energy Optimization:** Implement energy-efficient techniques for longer system runtime.

References

- <https://lastminuteengineers.com/esp32-arduino-ide-tutorial/>
- <https://randomnerdtutorials.com/esp32-dht11-dht22-temperature-humidity-sensor-arduino-ide/>
- <https://www.instructables.com/ESP32-MQ135-Gas-SensorAir-Quality-Index/>
- <https://www.electronicwings.com/esp32/bmp280-barometer-sensor-interfacing-with-esp32>
- <https://www.emqx.com/en/blog/esp32-connects-to-the-free-public-mqtt-broker>
- <https://randomnerdtutorials.com/esp32-esp8266-mysql-database-php/>



Thank You
